

FAST NUCES

Department of Software Engineering



BS in Software Engineering

Project Management Plan

**Intelligent System for Pakistan Sign Language
Learning and Translation**

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CHANGE HISTORY

Table 1: Change history

Version Number	Release Date	Pages Changed	Change Description
1.0	20 th September, 2025	1–12	Submitted version of the document

PREFACE

This document serves as template of the Project Management Plan (PMP) for the Ph.D. course Funding and Management of Research Projects (FMRP); reference books: [1, 2]. Note that it is intended for the Academic Year 2022/2023; the author is not responsible for any misuse should the course's regulations change in the future.

Feel free to reuse this template for your PMP, and good luck for your Funding and Management of Research Projects exam!

Here are the rules: do not panic, stay hydrated, and have fun!

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CHAPTER 1

PROJECT OVERVIEW

The “Intelligent System for Pakistan Sign Language Learning and Translation” will be a mobile application that combines computer vision, NLP, and deep learning to provide real-time PSL translation and structured PSL learning. It will include a PSL dictionary, lessons, and assessments to promote awareness, inclusivity, and accessibility.

1.1 Research Plan

1.1.1 Scope

The PSL Translation and Learning project will include the design, development, testing, and delivery of a mobile application that provides real-time Pakistan Sign Language (PSL) translation, an integrated dictionary, structured learning modules, and assessment features. All technical resources, including personnel, software frameworks, and database management systems, will be managed by the project team. Project activities will be carried out independently of daily academic and organizational operations, and all required testing will be conducted with pilot users and domain experts, including members of the Deaf Community. Project funding and resources will be managed by the project manager within the allocated limits defined in this document, with any additional requirements subject to sponsor approval. This project will be considered complete when the application is deployed on mobile platforms, all planned modules are functional, user testing and validation are successfully completed, technical documentation is finalized, and the final project report is submitted for approval.

1.1.2 Purpose

The purpose is to reduce the communication gap for the non-hearing community, provide an accessible learning platform for PSL, and promote inclusivity using modern AI technologies.

1.1.3 Objectives

- To develop a translation module for PSL-to-text/speech and text/speech-to-PSL.
- To create a structured learning module with lessons and practice sessions.

- To implement an assessment system with quizzes and sign replication tasks.
- To build a categorized PSL dictionary for easy reference.

1.2 Assumptions and Constraints

The project scope is limited to Pakistan Sign Language (PSL) only; support for other sign languages is out of scope for this phase. Resource costs mainly involve dataset preparation, time, and training resources.

It is assumed that sufficient PSL datasets will be available to train the AI models effectively and that users will have access to smartphones with reliable internet connectivity to use the application. The success of the system also depends on the cultural acceptance of PSL accuracy and representation by the Deaf community, as their validation is essential for credibility. Furthermore, it is assumed that collaborations with NGOs, schools, and institutions will support the sustainability and wider adoption of the project.

1.2.1 Personnel

The project will be managed by a student development team under faculty supervision. Roles include Project Manager, AI Developer, Mobile App Developer, and UI/UX Designer and tester. The roles will be carried by the three team members. Oversight and evaluation will be carried out by the Final Year Project (FYP) Jury, who will assess progress, provide feedback, and validate deliverables to ensure the project meets academic and technical standards.

1.2.2 Timing

The project is expected to be completed within 9-12 months, divided into dataset preparation, model training, module development, integration, testing, and final deployment.

1.2.3 Budget

Minimal costs are expected as the project relies on open-source tools and university computing resources. Additional costs may include cloud storage or GPU rentals for model training.

1.3 Activities and Deliverables

- Real-time PSL-to-text/speech and text/speech-to-PSL translation.
- Categorized PSL dictionary (alphabetical and chapter-wise).
- Learning module with lessons, tutorials, and exercises.
- Assessment system with quizzes, tests, and grading.
- Mobile deployment (Android) with user-friendly UI.
- Secure data storage ensuring user privacy.
- Thorough testing with stakeholders before full release.

CHAPTER 2

PROJECT CONTEXT

The “Intelligent System for Pakistan Sign Language Learning and Translation” project addresses the significant communication barriers between hearing and non-hearing individuals in Pakistan. Currently, there is a lack of accessible and affordable platforms that facilitate real-time Pakistan Sign Language (PSL) translation and structured learning. This project aims to fill that gap by providing a mobile application that not only translates PSL gestures into text and speech, but also enables users to learn and practice PSL through interactive tutorials, assessments, and gamified features.

The project is motivated by the growing social need for inclusivity, accessibility, and digital empowerment of the Deaf community. Leveraging artificial intelligence, computer vision, and natural language processing, the application will deliver real-time PSL gesture recognition and translation, supported by an engaging learning environment. The expected outcomes include increased awareness of PSL, improved accessibility for the Deaf community, and the promotion of inclusivity within Pakistani society.

2.1 Process Model

The project will follow an Iterative Process Model, ensuring continuous development, testing, and refinement of the system. Each iteration will focus on building core components such as the translation module, dictionary, learning, and assessment modules. Feedback from early prototypes will be used to improve subsequent iterations, ensuring that the final product meets user expectations and technical requirements.

2.2 Process Improvement Plan

To ensure continuous improvement, the project will adopt a feedback-driven enhancement cycle. Regular team meetings, user testing sessions, and evaluation metrics (such as accuracy, usability, and engagement) will guide refinements. Lessons learned from each iteration will be documented and applied to future cycles, reducing risks, improving efficiency, and enhancing the overall quality of the application.

2.3 Product Acceptance Plan

The product will be deemed acceptable upon meeting the following conditions:

- Accurate recognition and translation of gestures in Pakistan Sign Language (PSL).
- A functional learning module with lessons, practice tasks, and assessments.
- An intuitive and engaging mobile interface accessible to users of varying technical backgrounds.
- Compliance with defined success criteria, including cultural accuracy and user engagement.

2.4 Infrastructure, tools and methods

The project will leverage modern technologies and tools for efficient development:

- **Backend:** Python for model integration.
- **Frontend:** Flutter for cross-platform mobile application development.
- **Database:** MongoDB for storing user data, dictionary entries, and assessment results.
- **Machine Learning Libraries:** TensorFlow/Keras, PyTorch, Mediapipe, and Transformers for gesture recognition and NLP.
- **Version Control & Collaboration:** GitHub/Git for source code management and agile project tracking.
- **Testing & Deployment:** Firebase for hosting, authentication, and cloud storage; Docker for containerized deployment.

2.5 Project Organization

The project will be executed through a collaborative team structure:

- **Project Manager:** Responsible for overall planning, coordination, communication, and resource management.
- **Development Team:** Handles backend development, mobile application design, and database integration.
- **AI/ML Specialists:** Focus on dataset preparation, model training, gesture recognition, and NLP-based translation.
- **UI/UX Designers:** Ensure that the application interface is user-friendly, visually consistent, and engaging.
- **Quality Assurance Team:** Conducts testing of modules, validates functionality, and ensures compliance with requirements.
- **Stakeholders/Project Sponsor:** Provide approval, feedback, and guidance throughout the project lifecycle.

CHAPTER 3

PROJECT PLANNING

The project will be planned and executed in iterative phases, ensuring early testing and integration to reduce risks.

3.1 Work Activities

- **Phase 1 - Project Start:**

Official initiation of the project, kickoff meeting, and allocation of roles.

- **Phase 2 - Documentation:**

- (i) SRS (Software Requirements Specification).
- (ii) SDS (Software Design Specification)

- **Phase 3 - Core Modules Development:**

- (i) Authentication and Authorization Module (ii) Dictionary Module
- (iii) Translation Module

- **Phase 4 - Learning and Assessment:**

- (i) Learning Module
- (ii) Assessment Module

- **Phase 5 - Integration and Testing:**

Complete Solution Simulation and Testing

- **Phase 6 - Deployment:**

Deploy Solution

- **Phase 7 - Closure:**

Project Complete

3.2 Time Allocation

The project will span approximately nine months, beginning in September 2025 and concluding in June 2026. The time allocation is distributed as follows:

- **Project initiation and planning:** September 2025
- **Requirements and design documentation (SRS and SDS):** October to December 2025
- **Core module development (Authentication, Dictionary, Translation):** December 2025 to January 2026
- **Learning and Assessment modules:** February to March 2026
- **System integration and testing:** April 2026
- **Deployment:** May 2026
- **Final project completion and closure:** June 2026

CHAPTER 4

PROJECT ASSESSMENT AND CONTROL

The project's success depends on meeting core requirements, timely delivery, and maintaining quality standards. Progress will be tracked through milestones, regular meetings among developers and with the supervisor, and FYP evaluations carried out by the jury.

4.1 Requirements Management Plan

All requirements for the PSL Translation and Learning Application will be documented in the Software Requirements Specification (SRS) and validated with the faculty supervisor, project team, and FYP Jury. A requirements traceability matrix (RTM) will track each requirement across design, implementation, and testing to ensure completeness. Any changes will follow a formal change control process, with written requests analyzed for impact and approved by the supervisor. Major changes will also be reviewed by the FYP Jury. Requirements will be revisited at milestones (SRS, SDS, module completions, integration) to confirm progress before final deployment.

4.2 Scope Change Control Plan

Any proposed scope changes suggested by the jury will be formally documented. The Project Manager will review the impact of each request on timeline, resources, and deliverables, in consultation with the faculty supervisor. Changes affecting milestones or deliverables must also be presented to the FYP Jury for approval. Once approved, changes will be recorded in the project log, and the schedule and resource allocations will be updated accordingly. Unapproved changes will not be implemented to ensure project stability.

4.3 Schedule Control Plan

Project progress will be tracked weekly using a Gantt chart and milestone reviews. Delays will be identified early through progress meetings with the supervisor and team. In the event of slippage, the team will apply corrective actions such as redistributing tasks, extending work hours,

or adjusting scope in consultation with the supervisor. Significant schedule adjustments must be communicated to the FYP Jury during progress reviews.

4.4 Budget Control Plan

Although costs are minimal for this student project, any expenses (e.g., dataset curation, cloud services for model training) will be recorded in a budget tracking sheet. The Project Manager will ensure that expenses remain within available university or personal resources. Any additional funding requirements will be submitted to the supervisor for approval. Unplanned expenditures will only be authorized if they are critical to project success.

4.5 Quality Assurance Plan

The project will apply quality control measures throughout development to ensure reliability and usability:

- **Model Testing:** AI models will be evaluated using metrics such as accuracy, precision, recall, and F1-score. Multiple datasets (train/test/validation) will be used to verify robustness.
- **Application Usability:** The mobile application will undergo usability testing with target users (students, Deaf community representatives) to ensure intuitive navigation and effective interaction.
- **System Testing:** Functional, integration, and regression testing will be conducted before deployment.
- **Security and Privacy:** A privacy audit will ensure user data is stored securely, with no sensitive camera input saved unnecessarily.

4.6 Dissemination Plan

4.6.1 Scientific dissemination

The project results will be documented in the FYP dissertation and may be further developed into a research paper. Potential venues include AI and accessibility conferences or journals focusing on Computer Vision, NLP, and Assistive Technology. Collaboration with faculty may support co-authored publications.

4.6.2 General-purpose dissemination

Beyond academia, the project outcomes will be presented to the university community, NGOs, and schools for the Deaf. Workshops or awareness sessions may be conducted to promote PSL adoption and highlight the role of AI in accessibility. A demonstration of the application will also be shared during the FYP Final Jury presentations.

CHAPTER 5

SUPPORTING PROCESS PLANS

This chapter outlines the supporting processes that will guide the effective execution and sustainability of the PSL Translation and Learning Application. It provides a structured approach to identifying and managing risks, ensuring that both potential challenges and opportunities are recognized and addressed proactively. In addition, the chapter details cost management practices to ensure that resources are allocated efficiently, expenses remain within scope, and financial accountability is maintained throughout the project. Together, these plans support the long-term viability of the project and its successful implementation.

5.1 Risk Management

Effective risk management is essential to ensure the success and sustainability of the project. The risks identified fall into two categories: negative risks, which represent potential threats that can hinder adoption, performance, or scalability, and positive risks, which represent opportunities that can be leveraged to strengthen impact, improve accuracy, and ensure long-term viability.

5.1.1 Negative risks

The project is subject to several potential threats that can affect its success and sustainability.

From a political perspective, there is a possibility of limited acceptance if the application fails to gain the trust of the Deaf community or if stakeholders perceive it as culturally inaccurate.

Operationally, the effectiveness of the system could be compromised if Pakistan Sign Language (PSL) data sets are insufficient, inconsistently labeled, or inadequately tested, leading to reduced recognition accuracy.

Economically, the project may face challenges in scaling if sufficient external funding or NGO support is not secured, while the ongoing costs of hardware procurement and system maintenance may further constrain sustainability. On a technical level, the initial AI accuracy may be low due to the inherent complexity of gesture recognition and the limited availability of quality training data. In addition, restricted access to smartphones, Internet connectivity, and related resources among the intended user base presents a further barrier to adoption.

5.1.2 Positive risks

Despite these challenges, the project also presents a number of opportunities that can enhance its impact and long-term viability.

Politically, early engagement with representative organizations, coupled with validation by PSL experts, offers an opportunity to establish credibility and foster trust within the Deaf community. Collaboration with NGOs that already work closely with this community can further strengthen acceptance and cultural alignment.

Operationally, partnerships with universities and Deaf associations present an opportunity to enrich data set quality, improve testing processes, and ensure greater system reliability. Furthermore, an iterative development approach allows for continuous improvement of recognition accuracy over time.

From an economic and financial perspective, there are opportunities to secure sustainable funding through partnerships with NGOs, educational institutions, and grant-making organizations, while reliance on open-source tools reduces costs and enhances scalability potential. Technically, iterative model training, community-driven dataset expansion, and regular updates represent key opportunities to strengthen system performance and reliability, while ongoing community contributions can ensure adaptability and long-term relevance.

5.2 Cost Management

The PSL Translation and Learning Application will primarily rely on open source frameworks, university computing resources, and team effort, keeping direct financial costs minimal. Any potential expenses, such as dataset acquisition, cloud GPU services for model training, or mobile deployment costs, will be estimated in advance and documented in the project log.

Costs will be tracked using a simple budget tracking sheet, updated at each milestone. The Project Manager is responsible for monitoring expenses and ensuring that resources remain within the planned scope. Any additional costs must be approved by the faculty supervisor and, if significant, reported to the FYP Jury.

ACRONYMS

FMRP Funding and Management of Research Projects. ii

PMP Project Management Plan. ii

REFERENCES

- [1] Project Management Institute, ed. *A guide to the project management body of knowledge / Project Management Institute*. Sixth edition. PMBOK guide. Newtown Square, PA: Project Management Institute, 2017. ISBN: 978-1-62825-184-5.
- [2] Kathy Schwalbe. *Information technology project management*. Revised Seventh edition. OCLC: ocn864657391. Australia: Cengage Learning, 2014. ISBN: 978-1-285-84709-2.